BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 162 · Paper 2: Unseen Code and Debugging

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the raw mono sound size in bits?

Q2. What is a character set?
A. A list of fonts
B. Characters linked to codes
C. A list of filenames
D. A collection of pixel colours

Q3. Given code 65 for A and consecutive capital-letter codes, what is C's code?

Q4. A website needs English, Greek and Chinese text. “Unicode has fewer characters than standard ASCII.” Correct this. Why does Unicode suit this site?

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I Do · Read new knowledge

Key vocabulary

Read code literally — Do not assume it does what its title claims.

Trace state — Follow nested conditions and loops from initial values.

Identify fault location — Point to the exact condition, bound or assignment.

Repair and retest — State expected outputs for normal and boundary inputs.

New learning

Read code literally: Do not assume it does what its title claims.

Trace state: Follow nested conditions and loops from initial values.

Identify fault location: Point to the exact condition, bound or assignment.

Repair and retest: State expected outputs for normal and boundary inputs.

Worked example

Debug total=1 for a program intended to sum [2,4,6].

1. Trace initial total=1.

2. Add 2,4,6 → final 13, not 12.

3. Identify unwanted starting value.

4. Set total=0; expected result 12.

We Do · Practise with guidance

Guided 1

A password attempt counter locks after four failures instead of three. Fix it.

Worked method: Test attempts=2 and 3.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the loop bound so the largest file size is found.

```
array files = [6,3,10]
largest = files[0]
for i = 0 to 1
    if files[i] > largest then
        largest = files[i]
    endif
next i
print(largest)
```

iterate

Question 1

A loop runs i=0 to 4 inclusive. How many iterations execute? [2 marks]

Question 2

Explain why total=1 is wrong for a plain sum accumulator. [2 marks]

Question 3

Give a suitable test after fixing an upper boundary error. [3 marks]

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Exit Ticket · Answer before checking

Fixing an algorithm should be followed by testing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sample rate × duration in seconds × bits per sample.

Q2. B — Characters linked to codes

Q3. 67, because C is two positions after A.

Q4. Unicode supports far more characters than standard ASCII. It can cover English, Greek and Chinese. Use an encoding with enough codes; Unicode characters do not always take one byte.

I Do / We Do pages · Guided practice

Q1. Test attempts=2 and 3.

You Do pages · Code First

```
array files = [6,3,10]
largest = files[0]
for i = 0 to 2
    if files[i] > largest then
        largest = files[i]
    endif
next i
print(largest)
```

Expected result: 10

You Do pages · Independent questions

Q1. 5.

Q2. Adds an unintended 1 to every computed total; initialise to zero.

Q3. Test exactly at upper boundary and one beyond, plus normal value.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: